

KOKKOLA TANKAR, Finland

WMO: 027210

Lat: 63.951N

Lon: 22.844E

Elev: 21

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	0.5	-10.0	3.2	-5.7	-3.0	4.7	0.7	41.2	32.4	37.6	28.9	8.0	170	0.817

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	72.6	66.5	69.7	64.4	67.2	62.2	67.8	71.9	65.1	68.9	63.0	66.5	10.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	95.7	71.0	63.4	87.5	67.9	61.3	81.1	65.8	32.1	71.7	30.1	69.4	28.5	66.7	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
35.8	31.9	28.5	-6.9	77.9	7.6	4.8	-12.4	81.4	-16.8	84.2	-21.1	87.0	-26.7	90.5		
			-5.7	68.0	9.1	3.4	-12.2	70.4	-17.6	72.4	-22.7	74.3	-29.4	76.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	21.6	22.4	26.9	34.6	44.8	53.0	61.0	59.8	53.2	42.3	35.2	29.0	
	DBStd	15.19	10.16	10.86	7.80	4.60	5.89	5.29	5.51	4.65	4.57	5.12	5.82	8.57	
	HDD50	4401	882	774	717	461	183	25	0	1	23	240	444	651	
	HDD65	9011	1347	1194	1182	911	627	360	152	172	353	703	894	1116	
	CDD50	905	0	0	0	0	21	116	340	305	120	3	0	0	
	CDD65	41	0	0	0	0	0	2	27	12	0	0	0	0	
	CDH74	65	0	0	0	0	0	3	32	30	0	0	0	0	
	CDH80	7	0	0	0	0	0	0	1	6	0	0	0	0	
Wind	WSAvg	14.1	15.1	14.5	14.6	13.1	13.4	13.1	11.6	12.6	13.6	15.7	15.8	16.8	
Precipitation	PrecAvg	17.8	1.2	0.9	0.9	0.9	1.3	1.6	2.3	2.0	2.0	1.9	1.7	1.4	
	PrecMax	23.8	2.4	1.9	1.9	2.4	2.5	3.3	4.6	5.4	4.1	3.4	4.0	3.0	
	PrecMin	12.4	0.1	0.0	0.0	0.1	0.0	0.2	0.0	0.6	0.8	0.4	0.0	0.2	
	PrecStd	3.1	0.6	0.6	0.5	0.6	0.7	0.9	1.3	1.3	1.1	0.8	0.9	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	39.3	43.1	48.6	65.1	70.2	77.4	78.3	66.6	56.4	48.4	41.7	
		MCWB	37.1	36.9	37.5	44.3	56.0	60.8	69.2	68.0	60.7	52.6	46.5	40.2	
	2%	DB	36.4	37.1	39.3	45.4	59.1	66.0	74.0	71.4	62.6	52.5	46.4	40.3	
		MCWB	35.1	35.2	36.2	41.4	51.9	59.4	67.6	65.1	58.5	50.0	44.8	38.8	
	5%	DB	34.8	35.7	37.5	42.8	55.4	62.9	71.1	68.7	60.6	50.5	44.2	38.9	
		MCWB	33.7	34.2	35.2	39.1	50.0	58.1	66.0	63.0	57.0	48.3	43.0	37.5	
	10%	DB	33.5	34.1	35.9	40.7	52.7	60.4	68.8	66.0	59.0	49.1	42.3	37.3	
		MCWB	32.6	33.0	33.9	37.7	48.0	56.1	64.2	61.2	56.0	47.0	40.9	35.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	37.5	38.6	45.6	56.8	62.9	70.7	70.0	61.1	53.1	46.8	40.8
			MCDB	38.3	38.6	41.1	47.3	63.2	67.7	75.0	76.2	64.8	54.8	47.9	41.1
2%		WB	35.4	35.6	36.7	41.9	53.0	60.5	68.8	66.5	59.3	50.4	44.9	39.2	
		MCDB	36.1	36.8	38.6	45.1	58.4	64.3	73.1	71.1	61.9	51.9	46.0	40.2	
5%		WB	34.0	34.4	35.4	39.6	50.6	58.5	66.8	63.7	57.8	49.0	43.1	37.7	
		MCDB	34.7	35.5	37.2	42.2	54.7	62.6	70.8	67.6	60.1	50.4	44.1	38.7	
10%		WB	32.8	33.2	34.1	38.1	48.5	56.6	64.6	62.0	56.5	47.3	41.2	36.2	
		MCDB	33.6	34.1	35.6	40.4	52.0	60.1	68.0	65.2	58.6	48.8	42.1	37.1	
Mean Daily Temperature Range		5% DB	MDBR	8.0	7.8	8.4	7.0	8.1	7.2	6.5	6.8	5.8	5.2	5.0	5.8
			MCDBR	5.4	5.9	7.7	10.2	13.6	11.1	9.8	10.7	8.0	6.4	5.6	6.1
	MCWBR		5.3	5.3	5.9	7.0	8.4	6.8	5.9	5.8	5.5	5.6	5.2	5.6	
	5% WB	MCDBR	5.9	6.2	7.0	9.5	12.6	9.9	8.8	9.4	7.5	6.5	5.6	6.0	
		MCWBR	5.8	5.6	5.7	7.0	8.6	6.5	6.0	6.0	5.7	5.9	5.4	5.7	
Clear-Sky Solar Irradiance	taub		0.225	0.286	0.311	0.350	0.350	0.351	0.371	0.358	0.333	0.314	0.358	0.351	
	taud		2.013	2.142	2.194	2.251	2.442	2.485	2.454	2.483	2.528	2.371	2.592	2.567	
	Ebn at Noon		110	196	244	261	274	275	266	259	242	190	102	53	
	Edh at Noon		9	20	28	34	31	30	30	27	21	16	8	5	
All-Sky Solar Radiation	RadAvg		41	221	704	1221	1650	1830	1666	1264	717	264	58	17	
	RadStd		4	44	77	108	161	155	153	117	74	48	10	1	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (45 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page